2002 Recert

ABT RECERTIFICATION EXAMINATION #19

May 5, 2001 ***PART A***

This examination constitutes part of the recertification process for toxicologists certified by the American Board of Toxicology in 1993 and 1998. There are three parts in this exam and an answer sheet is provided for each part. These answer sheets must be returned to ABT, P.O. Box 30054, Raleigh, NC 27622 for grading. An analysis of your performance in each part of the exam will be provided to the recertification committee of ABT and they will contact you regarding this and other requirements of the recertification process. If you have any questions please call the ABT Office at 919-572-5525.

EXAMINATION CATEGORIES

1. Metals, solvents, pesticides, gases, dusts and aerosols, industrial chemicals and radiation, plant and animal toxins, food additives, drugs and cosmetics.
2. Mutagenesis, carcinogenesis, developmental and reproductive toxicology, pulmonary toxicology, neurologic toxicology, immunologic toxicology, cutaneous toxicology, ocular toxicology, hepatic toxicology, hematopoietic toxicology, renal toxicology.
3. General principles, toxicokinetics, toxicodynamics and mechanisms of action, risk assessment, epidemiology and biostatistics, regulatory toxicology and risk management, environmental toxicology, occupational toxicology, forensic and clinical toxicology.

INSTRUCTIONS

1. Use a soft (#2) pencil to fill the box corresponding to your answer.
2. Select only one answer for each question. (Erase others)
3. Do not leave any questions unanswered. (No penalty for guessing).
4. Sign answer sheet ONLY on line that says name.
5. Do not bend or staple answer sheets.

Good luck!

Key for the following questions
Choose the single most appropriate answer

1. The major consequence of inhalation exposure to beryllium in an industrial setting is
   A) damage to the liver, with increased serum enzyme activity
   B) neurological symptoms, such as tingling and numbness in the extremities
   C) anemia
   D) gastrointestinal symptoms such as nausea and intestinal cramps
   E) respiratory dysfunction due to pulmonary fibrosis

2. The major mode of energy transfer leading to biological damage from ionizing radiation is
   A) spallation (breaking apart) of ^{12}C nuclei
   B) excitation of DNA atomic electrons
   C) electron-positron pair production
   D) electron capture
   E) ion pair production

3. Inhalation exposure to which of the following forms of silica (SiO_2) is most likely to lead to pulmonary fibrosis?
   A) amorphous silica
   B) quartz dust
   C) diatomaceous earth
   D) silica gel
   E) none of the above

4. Hydrogen sulfide is believed to act by a similar mechanism to what other toxicant?
   A) sulfuric acid mist
   B) cyanide
   C) sodium nitrite
   D) carbon monoxide
   E) none of the above

5. Which of the following long-term occupational exposures may lead to the development of neurological symptoms resembling those of Parkinson's disease?
   A) manganese in welding fumes
   B) lead in manufacture of lead-based pottery
   C) cobalt in manufacture of carbide-tipped tools
   D) hexavalent chromium in chrome plating
   E) cadmium in electroplating of steel

6. Signs of sudden vomiting and diarrhea occurred in an individual after oral ingestion of an unknown toxic substance. Fifteen days later he had muscle pain in his legs followed by progressive muscle weakness over few days. Recovery of muscle function was slow over a period of months. Which of the following is the most likely cause of his symptoms?
   A) phenobarbital
   B) carbolic acid (phenol)
   C) tri-*o*-cresylphosphate
   D) carbon tetrachloride
   E) mercuric bichloride

7. In methanol poisoning the most likely cause of eye pathology is
   A) acetaldehyde
   B) acetone
   C) methanol
   D) retinol
   E) formic acid

8. Which of the following is NOT characteristic of chelation therapy for heavy metal poisoning? The chelating agent
   A) competes with binding sites in the body for free metal ions
   B) converts the metal to a different chemical form
   C) makes the metal more water-soluble
   D) enhances the excretion of metal from the body
   E) may remove some essential metals for the body

9. Which of the following associations between exposure to a metal and a toxic effect in humans is INCORRECT?
   A) arsine - hemolytic anemia
   B) beryllium - granulomata in the lung
   C) cadmium - damage in distal renal tubules
   D) hexavalent chromium - perforation of nasal septum
   E) nickel carbonyl - acute gastrointestinal and respiratory tract symptoms

10. In adult humans the acute oral dose of DDT required to cause noticeable signs of toxicity is
   A) less than a microgram
   B) less than a milligram
   C) approximately 100 milligrams
   D) less than a gram
   E) more than 1 gram

11. The toxic action of plants containing high concentrations of oxalic acid includes
   A) hypocalcemia and kidney damage
   B) copper deficiency and liver damage
   C) molybdenum deficiency and hair loss
   D) zinc deficiency and lung damage
   E) hypophosphatemia and brain damage

12. Metabolism of which of the following generates carbon monoxide?
A) ethylene dibromide
B) methanol
C) methylene chloride
D) methylene glycol
E) hexane

13. Which of the following is true concerning amitrole?
A) has a very low acute toxicity, and no residue buildup when properly applied; however, its use was restricted for poultry intended for human consumption due to mutagenic activity
B) is goitrogenic in laboratory animals and is the agent involved in the cranberry crisis of 1959
C) is a dithiocarbamate fungicide/miticide used to fumigate infested poultry raising facilities and like other dithiocarbamates degrades to ETU, a goitrogen. Its use was restricted due to egg residues in 1968
D) is teratogenic in many species, including primates, and is the agent involved in the Texas grain episode
E) none of the above

14. The primary lethal effect of a rodenticide containing 2-fluoroacetamide is
A) general inhibition, by fluoride, of enzymes including lipases, esterases and phosphatases
B) direct inhibition of the synthesis of acetyl coenzyme A and blockage of energy metabolism
C) metabolism to fluorocitrate and Kreb's cycle blockage
D) binding with the acetylcholine receptor of the postsynaptic membrane followed by blockage of neurotransmission
E) interference with state 3 and state 4 respiratory activity by virtue of interaction with cytochrome oxidase resulting in inhibition of electron transport

15. The initial interaction between ionizing radiation and a cell component that results in cell damage or death is between the radiation and
A) H_2O, resulting in the generation of free radicals
B) RNA, resulting in the disruption of protein synthesis
C) DNA, resulting in the formation of thymine dimers
D) the cell membrane, resulting in changes in ion fluxes
E) polysomes, resulting in inhibition of protein synthesis

16. Repeated exposures over the course of several days are typically required to elicit the toxic effects of which of the following pesticides?
A) 2,4-dicholorophenoxyacetic acid
B) paraquat
C) parathion
D) permethrin
E) warfarin

17. Dilated pupils occurring in farm workers over a period of days is known as "cornpicker's pupil." It is caused by exposure to parasympatholytic substances from which of the following plants?
A) mountain lupine (*Lupinus formosus*)
B) sheep's sorrel
C) jimson weed (*Datura stramonium*)
D) meadow saffron (*Colchicum autumnale*)
E) pokeweed (*Phytolacca americana*)

18. Which of the following organisms produces ciguatoxins?
A) cobra snake
B) jellyfish
C) bacteria
D) benthic dinoflagellate
E) toads

Key for the following questions
Choose A) if 1,2 and 3 are correct
Choose B) if 1 and 3 are correct
Choose C) if 2 and 4 are correct
Choose D) if only 4 is correct
Choose E) if all four answers are correct

19. Which of the following statements is/are correct?
1) inhalation of cadmium fumes may produce an acute chemical pneumonitis and pulmonary edema
2) the principal, long-term effects at low-level exposure to cadmium are chronic obstructive pulmonary disease, emphysema and chronic renal tubular disease
3) the lung and renal disease produced by cadmium may be accompanied by effects on the cardiovascular and skeletal systems
4) chelation treatment with BAL can be used in the treatment of chronic cadmium exposure because the cadmium is bound to metallothionein

20. Which of the following statements is/are true for both dimercaptopropanol (BAL) and edetic acid (EDTA)?
1) act by chelating metals
2) can be toxic
3) effective antidote for lead poisoning
4) rapidly metabolized

21. Organophosphates used as therapeutic agents in veterinary medicine
1) can be applied by dipping, spraying, pouring-on and orally
2) may require metabolic activation
3) are of concern in regard to residues which may contaminate animal products destined for human consumption
4) have a large therapeutic index

22. Which of the following statements is/are correct regarding the toxicity of thallium in humans?
1) alopecia may occur 1-3 weeks after exposure
2) thallium poisoning may induce psychological and emotional changes
3) thallium can replace potassium in the NaK-ATPase pump
4) EDTA is an effective chelator of thallium

23. Benzene metabolites that appear in the urine include
1) ethereal sulfates
2) glucuronides of the phenolic metabolites
3) muconic acid resulting from ring opening
4) mercapturic acid resulting from glutathione conjugation

24. Potential benzene metabolites are
1) hydroquinone
2) catechol
3) o-benzoquinone
4) toluene

25. Dispositional tolerance (decreased amount of effective toxicant reaching the target site) is associated with repeated exposure to which of the following metals?
1) nickel
2) barium
3) antimony
4) cadmium

26. Dietary selenium compounds modify the toxicity of which of the following?
1) arsenic
2) cadmium
3) mercury
4) silver

27. Which of the following statements about ethylene dibromide (EDB) is/are correct?
1) antispermatogenic effects of EDB have been observed in occupationally exposed humans
2) oral administration of EDB causes granular deposits in the corneal endothelium
3) EDB was used as a pesticide and grain fumigant in the 1970's
4) EDB breaks down rapidly in soil and ground water

28. The fumigant methyl bromide
1) is highly lipophilic and crosses the blood-brain barrier readily
2) is severely irritating to the skin and to the lower respiratory tract causing skin burns, pulmonary edema and hemorrhage
3) causes CNS effects including headache and visual disturbances
4) death from this agent is caused by depression of the CNS

29. Which of the following statements is/are true about ambient air pollutants?
1) SO_2 is a water soluble gas that is an upper airway irritant and causes bronchoconstriction
2) ozone is a poorly soluble, deep lung irritant that damages Type I cells and causes inflammation in the lungs
3) aerosolized sulfuric acid increases airway reactivity, especially in asthmatic humans and guinea pigs
4) NO_2 isa water-soluble, deep lung irritant that damages Type I cells

30. Pulmonary disease has been associated with occupational exposure to dusts and fibers. Which of the following statements is/are true about pneumoconiotic agents?
1) silicotic nodules formed by protein encapsulation characteristically appear as "dumbbell" shaped structures in the lung
2) Farmer's Lung may be characterized as extrinsic allergic alveolitis
3) asbestos exposure is associated with emphysema and increased susceptibility to tuberculosis
4) alveolar macrophages are important in the toxic responses to asbestos and silica exposures

31. Anatoxin A
1) is a cause of sudden death in livestock
2) produces a neuromuscular blockade
3) is produced by a species of blue-green algae
4) is an inhibitor of the release of acetylcholine from presynaptic vesicles

32. Oxygen toxicity can result in which of the following conditions?
1) adult respiratory distress syndrome (diffuse alveolar damage)
2) bronchopulmonary dysplasia in infants
3) retrolental fibroplasia (retinopathy) in infants
4) free radical damage to alveolar capillary walls

33. Deposition of inhaled particles may occur by
1) diffusion
2) interception
3) sedimentation
4) impaction

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34. Which of the following represent(s) correct ordering for sensitivity (more sensitive to less sensitive) of cells to the adverse effects of ionizing radiation?
1) differentiating intermitotic cells - differentiating postmitotic cells - terminally differentiated cells
2) stem cells - differentiating cells - differentiated specialized cells
3) hematopoietic stem cells - oocytes - neurons
4) erythrocytes - mesenchymal cells - hematopoietic stem cells

35. Terfenadine can produce lethal cardiotoxic effects if administered with
1) azole antifungals
2) penicillin
3) macrolide antibiotics
4) antivirals

36. Common adverse effect(s) of alkylating agents and antimetabolites is/are
1) lymphocytopenia
2) neutropenia
3) anemia
4) thrombocytopenia

37. Phenobarbital (PB) has been associated with thyroid tumors. Which of the following is/are true about PB and thyroid tumors
1) PB is a tumor promoter
2) PB is a direct acting carcinogen
3) PB induces hepatic microsomal enzymes that cause disruption of thyroid function through increased metabolism of T_4, leading to thyroid tumors in rats
4) there is convincing evidence that humans treated with PB are at increased risk for thyroid tumors

38. Heparin is the preferred anticoagulant for use in pregnant women because
1) it does not cross the placenta
2) it can be administered orally
3) it is not known to produce skeletal malformations in offspring
4) there are no adverse effects on pregnancy outcome

39. Mycotoxins represent a diverse group of mold species-specific metabolites that can occur in a variety of plant and animal products. Which of the following associations of mycotoxin with target organ toxicity is/are correct?
1) tricothecenes - gastrointestinal toxicity
2) zearalenone - reproductive organ toxicity
3) ochratoxins - renal toxicity
4) ergot - vascular toxicity

40. The rat lung responds to chronic inhalation exposure to high levels of inert particulate matter by developing tumors. Which of the following is/are correct?
1) similar tumors are not observed in mice similarly exposed
2) tumors appear to arise via an "overload" mechanism that reduces particle clearance leading to chronic inflammation and DNA damage in lung epithelial cells
3) the particulate matter responsible includes diesel exhaust, carbon black and titanium dioxide
4) the particulate matter must be of uniform size to cause lung tumors

41. Which of the following mycotoxins are suspect carcinogens based on human or animal data?
1) ochratoxin
2) fumonisin
3) aflatoxin B_1
4) sterigmatocystin

42. Which of the following is/are true regarding amygdalin?
1) it is a source of cyanide poisoning
2) it produces free radicals that bind with corneal epithelium to cause corneal opacity
3) it requires activation by intestinal microflora to exert its toxicity
4) it is associated with liver cirrhosis

43. Alkyl benzenes such as toluene and xylene
1) exert their acute effects primarily through CNS depression
2) are mutagenic in the Ames *Salmonella* reverse mutation assay with or without S9 activation
3) are metabolized to readily excretable metabolites which have a low order of toxicity even at relatively high doses
4) have been shown to cause liver toxicity and hepatic cancer following chronic exposure

44. Food allergy is
1) often a reaction to carbohydrates in food
2) mediated by the immune system
3) more common in adults than children
4) often confused with food intolerance or other adverse reactions to food

45. Which of the following is/are true?
1) trichloroethylene is mainly metabolized by the P-450 family of enzymes in the liver
2) trichloroethanol and trichloroacetic acid have been shown to induce peroxisomes in rodent livers
3) at high doses, metabolism of trichloroethylene is saturable in rats and mice
4) the major metabolites of trichloroethylene in mice, rats and humans are trichloroethanol and trichloroacetic acid

46. Toxic exposure to which of the following industrial solvents is likely to cause peripheral neuropathy?
   1) carbon disulfide
   2) hexane
   3) methyl butylketone
   4) dioxane

47. Which of the following statements is/are true regarding the safety evaluation of direct food additives or color additives used in food as reflected in FDA's Toxicological Principles for the Safety Assessment of Direct Food Additives and Color Additives Used in Food?
   1) exposure is a key factor in an algorithm used to determine minimum testing requirements
   2) there must be "reasonable certainty of no harm under the conditions of intended use"
   3) the Estimated Daily Intake is a measure of the expected human intake (mg/day) based on the intended use level of the additive and the daily consumption levels of the food products containing the intended food additive
   4) it must be demonstrated that the food additive has no toxicity by the intended route of exposure

48. Under the Federal Food, Drug and Cosmetic Act, safety must be demonstrated prior to marketing for all of the following EXCEPT
   1) GRAS substances
   2) indirect food additives
   3) prior sanctioned ingredients
   4) direct food additives

49. Which commonly consumed foods contain phytoestrogens that could interfere with hormonal activity in animals and humans?
   1) wheat
   2) flax seed
   3) corn
   4) soy

50. The acute-phase response is a systemic reaction to inflammation or tissue injury and is often observed with cytokine administration. It is characterized by
   1) leukocytosis
   2) fever
   3) increased vascular permeability
   4) increased levels of acute-phase proteins

Key for the following questions:
Choose the single most appropriate answer

1. Which of the following is NOT true regarding erythropoietin?
   A) is produced by the liver in the fetus and by the kidney in adults
   B) stimulates leukocyte production
   C) production is stimulated by hypoxia
   D) primary site of action is the bone marrow
   E) Fe^{59} has been used to assay cellular proliferation caused by erythropoietin

2. Belladonna antimuscarinic agents are used predominantly as selective parasympatholytics. Local administration of these agents to the eye produces
   A) mydriasis
   B) conjunctivitis
   C) cycloplegia
   D) A and B
   E) A and C

3. Biliary canaliculi
   A) carry bile toward the portal triads
   B) contain microvillar insertions from Kupffer cells
   C) can become clogged with cell debris from disintegrated bile duct epithelium
   D) A and C
   E) none of the above

4. Which of the following is poorly absorbed through the skin?
   A) sarin
   B) DDT
   C) diazinon
   D) parathion
   E) aldicarb

5. An agent regularly associated with pancytopenia is
   A) toluene
   B) benzene
   C) xylene
   D) epichlorohydrin
   E) ethylene oxide

6. Which of the following adverse reproductive outcomes of exposure to environmental toxicants is most likely to be detected in humans?
   A) infertility in the male and abortion in the female
   B) Sertoli cell tumors in the male and premature labor in the female
   C) Leydig cell tumors in the male and intrauterine fetal growth retardation
   D) altered libido in both males and females
   E) abnormal sperm production and dysmenorrhea

7. Excessive loss of prepubertal follicles in the human female may result in
   A) malformed offspring
   B) early onset of menopause
   C) precocious puberty
   D) no change in fertility, since follicles are continually replaced
   E) no change in fertility, since far more follicles are formed than could ever be ovulated during an average woman's reproductive life time

8. Men may be more sensitive to spermatotoxic agents than nonprimate mammalian species because men
   A) produce fewer sperm per gram of testis
   B) have a higher incidence of abnormal sperm
   C) have smaller epididymal sperm reserves
   D) copulate with greater frequency
   E) copulate with lesser frequency

9. Sperm production rate can be assessed in reproduction toxicity studies by
   A) testes weight
   B) epididymal sperm numbers
   C) sperm motility
   D) sperm concentration in vaginal lavage
   E) sperm transit times

10. The primary site of aminoglycoside antibiotic nephrotoxicity is the
   A) cortico-medullary junction
   B) proximal tubule
   C) loop of Henle
   D) distal tubule
   E) collecting duct

11. The blood-brain barrier consists of
   A) myelin, which forms a lipophilic barrier around the neuron
   B) an ATP-dependent "pumping" system, which actively transports xenobiotics out of the neuron
   C) endothelial and glial cells
   D) tight junctions between oligodendrocytes
   E) the meningial membranes of the central nervous system

12. Which of the following is a powerful vasodilator of cerebral vessels?
A) carbon dioxide
B) helium
C) oxygen
D) nitrogen
E) all of the above

13. Theophylline blocks
A) antigen formation
B) mast cell degranulation
C) cyclic AMP degradation
D) estrogen receptor binding
E) antibody production

14. A reduction of lymphoid follicles and germinal centers in the spleen is indicative of
A) T-cell deficiency
B) neutropenia
C) B-cell deficiency
D) A and C
E) none of the above

15. Which of the following statements is/are correct?
A) cross-tolerance is observed between O_3 and NO_2
B) atropine blocks SO_2 bronchoconstriction
C) SO_2 increases air flow resistance and decreases elasticity
D) A and C
E) all of the above

16. Which of the following compounds does/do NOT cause fatty liver or centrilobular necrosis?
A) methylene chloride
B) chloroform
C) carbon tetrachloride
D) kerosene
E) A and C

17. What is the major fate of bilirubin excreted into the intestine?
A) conversion to urobilinogen by the flora of the large intestine
B) reabsorption and recirculation to the liver
C) conversion to bile salts by pancreatic enzymes
D) reabsorption and excretion by the kidney
E) esterification and reabsorption

18. Thermophilic actinomycetes are frequently the antigenic agents responsible for the disease called
A) black lung
B) farmer's lung
C) polymer fume fever
D) ornithosis
E) none of the above

19. Haber's law states that
A) exposure chamber gas concentration is the product of a constant and the duration of exposure
B) mortality is directly related to exposure chamber gas concentration
C) the product of the exposure chamber gas concentration and time of exposure required to produce a specific physiologic effect is equal to a constant
D) the administered dose of gas can be controlled within a specified concentration range and chamber airflow conditions
E) none of the above

Key for the following questions
Choose A) if 1,2 and 3 are correct
Choose B) if 1 and 3 are correct
Choose C) if 2 and 4 are correct
Choose D) if only 4 is correct
Choose E) if all four answers are correct

20. Which of the following statements is/are true regarding the estrous cycle in the rat?
1) the cycle stage can be determined by examining vaginal smears
2) a predominance of cornified cells in the vagina indicates a high level of estrogen and a predominance of leukocytes in the vaginal wall indicates a high level of progesterone
3) the viscosity of cervical mucus changes during the estrous cycle
4) copulation usually occurs during diestrus

21. Nitric acid vapor inhalation can result in
1) pulmonary edema
2) bronchial asthma
3) pulmonary fibrosis
4) hepatic necrosis

22. Pregnancy alters several factors that influence the distribution of xenobiotics including
1) increase in total body water
2) increase in plasma binding proteins
3) elevated plasma volume
4) decrease in body fat

23. Which of the following agents partition(s) in the lipid compartment of neuronal membranes and produce(s) CNS depression?
1) ethyl alcohol
2) nitrous oxide
3) halothane
4) DDT

24. Antisense oligonucleotides can interfere with the developing embryo by
1) ablating multiple gene family members
2) causing temporal and spatial restriction of gene ablation
3) inducing a loss of gene function for the duration of development
4) hybridization with cellular mRNA

25. Which of the following statements is/are true regarding excitatory neurotoxicity?
1) the effects of excitotoxicants preferentially affect axons of the CNS
2) the entry of glutamate into the brain occurs by passive diffusion across the blood-brain barrier
3) following exposure to an excitotoxic chemical, a 7-10 day "silent" period precedes the expression of clinical signs
4) toxicity of kainate and domoic acid are mediated by excitotoxic mechanisms

26. Which of the following compounds has/have been associated with teratogenic effects when administered to humans during the first trimester of pregnancy in therapeutic doses?
1) oral contraceptives
2) isotretinoin
3) Bendectin® (doxylamine and pyridoxine)
4) thalidomide

27. Argyria, a bluish pigmentation of the skin, may be produced by
1) ingestion of trivalent arsenic
2) the prolonged treatment with gold-containing medicines
3) exposure to small amounts of mercury over extended periods of time
4) industrial exposure to silver

28. Transgenic Tg.AC mice typically develop which of the following tumors in response to topical application of a tumor promoter or carcinogen?
1) malignant lymphoma
2) lung adenoma
3) hepatocellular carcinoma
4) squamous cell papilloma

29. Mutagenicity tests of concentrated urine can indicate mutagen exposure in epidemiological studies; however, life-style habits, occupational exposures, or concurrent drug intake may confound study interpretation. Which of the following is/are confounding variable(s)?
1) cigarette smoking
2) consumption of fried foods
3) therapy with metronidazole
4) occupational exposure to asbestos

30. Of the four general types of hypersensitivity responses, which involve(s) serum antibody?
1) Type I
2) Type II
3) Type III
4) Type IV

31. Complement is involved in which of the following classes of reactions?
1) reagin dependent
2) cytotoxic
3) tuberculin type
4) damage by antigen-antibody complexes

32. Which of the following is/are true concerning the percutaneous absorption of chemical substances?
1) permeability is usually inversely proportional to the thickness of the stratum corneum
2) absorption is enhanced by hydration of the stratum corneum
3) in preterm neonates, the permeability of the skin is increased
4) the permeability to corticosteroid treatment is significantly increased in children with severe dermatitis

33. Which of the following statements about lacrimators is/are true?
1) lacrimators in threshold concentrations cause instant sensation but no tissue damage
2) mustards show a long latent period of subjective symptoms and then severe damage
3) lacrimators in high concentrations cause chemical burns with loss of corneal epithelium
4) alpha-chloroacetophenone and ethyl iodoacetate are common lacrimators

34. Histamine
1) contracts smooth muscle of the bronchi
2) stimulates gastric acid secretion
3) relaxes smooth muscle of small blood vessels
4) causes blurring of vision

35. Generally, low doses of agents that are rodent thyroid carcinogens at high doses do not pose a carcinogenic risk because
1) exposures are not high enough to decrease circulating T_4 levels
2) table salt is fortified with iodine
3) the chemicals display a threshold for causing thyroid tumors
4) binding to estrogen receptors decreases the chemicals' interactions with DNA

36. Which of the following statements is/are true about the events or agents associated with the known stages of carcinogenesis?
1) modulating factors such as hormones and diet can influence promotion
2) progressor agents may induce chromosomal aberrations and enhance clastogenic activity associated with karyotypic instability
3) alterations of xenobiotic metabolism, DNA repair or cell proliferation may influence initiation
4) promoting agents irreversibly enhance programmed cell death

37. Anemia in the laboratory rat can be associated with
1) a less than normal concentration of red blood cells and hemoglobin in the blood, and a decreased hematocrit
2) extamedullary hematopoiesis in the spleen
3) extramedullary hematopoiesis in the liver
4) an increase in reticulocytes in the blood

38. Which of the following may result from biliary excretion of a chemical?
1) increased hepatic exposure due to enterohepatic circulation
2) hepatic damage due to saturation of the biliary pathway
3) production of toxic metabolites in the gastrointestinal tract due to contact with gut microflora
4) increased likelihood of toxicity to gastric epithelium

39. Which of the following statements about the causes of cancer is/are true?
1) diet is among the most important risk factors for stomach, colon, and certain other cancers
2) inheritance of the gene BRCA1 and BRCA2 accounts for the vast majority of breast cancer cases in the US
3) human papilloma viruses are the major cause of cervical cancer in young women
4) general environmental pollution of air and drinking water is a relatively more important contributor to total cancer etiology than is cigarette smoking

40. The International Agency for Research on Cancer (IARC) has concluded that there is sufficient evidence of carcinogenicity in humans (Group 1 Carcinogen) for the following
1) asbestos
2) benzene
3) aflatoxins
4) 2-naphthylamine

41. Which of the following can modulate chemical carcinogenesis in laboratory rodents?
1) species
2) age at carcinogen exposure
3) micronutrient content of diet
4) animal husbandry

42. Which of the following is/are true?
1) 2,4-dinitrophenol produces cataracts in humans and rabbits, but not in rats or mice
2) busulfan exerts its cataractogenic effects by interfering with cell division in the lens epithelium
3) triparanol produces cataracts by causing a large increase in lens sodium content which results in hydration and swelling
4) cataract formation by galactose and other sugars can be prevented by treatment with aldose reductase inhibitors

43. Nephrotoxicity caused by carbon tetrachloride or chloroform is characterized by
1) polyuria
2) glucosuria
3) proteinuria
4) anuria

44. Which of the following statements is/are correct?
1) a compound that undergoes rapid detoxification in the liver may be more toxic when administered parenterally as compared to orally
2) since the major metabolic pathway for acetaminophen is via P450 metabolism and conjugation with glutathione, the rates of hepatic glucuronidation or sulfation have little effect on acetaminophen toxicity
3) although hepatic phase I enzymes can increase the toxicity of compounds by making them more reactive, phase II enzymes usually result in more stable and less toxic metabolites
4) chemically induced liver toxicity is rare because the liver has a great capacity to detoxify compounds

45. Which of the following conditions may require iron chelation therapy?
1) liver cirrhosis
2) hemochromatosis
3) chronic nephropathy
4) transfusional siderosis

46. Bromobenzene is
1) both a hepatotoxicant and a nephrotoxicant
2) activated to a reactive intermediate in the liver
3) metabolized in rats to 2-bromohydroquinone and covalently binds to macromolecules
4) metabolized in mice to ortho-, and para-bromophenols as well as 4-bromocatechol, all of which are nephrotoxic *in vivo*

47. Chemicals known to cause an increased incidence of renal tumors in male rats include
1) bromodichloromethane
2) trichloroethylene
3) chloroform
4) 1,4-dichlorobenzene

48. The kidney is susceptible to injury from toxicants for which of the following reasons?
1) the kidneys receive 20-25% of the cardiac output
2) the process of forming urine may concentrate toxicants and drive toxicants into tubular cells by diffusion
3) concentration of urine may precipitate relatively insoluble compounds in the tubules
4) bioactivation of toxicants can occur in the proximal tubules

49. A carcinoma
1) metastasizes only to regional lymph nodes
2) usually metastasizes first to the lungs
3) is highly undifferentiated
4) is a malignant epithelial tumor

50. Which of the following is/are likely to be involved in the etiology of lung cancer by cigarette smoke?
1) mutagenesis
2) immunosuppression
3) cocarcinogenesis
4) tumor promotion

Key for the following questions
Choose the single most appropriate answer

1. If dose-response data for a chronic irreversible effect of a chemical are available at relatively high dose levels
   A) it is generally possible to estimate the percentage response at very low dose levels by linear extrapolation of the log dose-response curve
   B) extrapolation to effects at low dose levels is highly uncertain since the shape of the dose-response curve at these levels is generally unknown
   C) it is generally possible to extrapolate the dose-response curve to the zero response intercept to obtain a precise threshold dose for the effect
   D) none of the above
   E) all of the above

2. Which of the following is NOT a common site of blood collection for the rat?
   A) retroorbital sinus
   B) ear vein
   C) heart
   D) aorta
   E) tail

3. The total plasma concentration of Compound X is determined to be 16 µg / 100 ml. The chemical is approximately 50% bound to the plasma protein. The steady state concentration of Compound X in the cerebrospinal fluid, assuming Compound X passes the blood brain barrier, would be expected to be approximately
   A) 0 µg / 100 ml
   B) 2 µg / 100 ml
   C) 4 µg / 100 ml
   D) 8 µg / 100 ml
   E) 16 µg / 100 ml

4. Hydroxylation of a chemical usually produces a
   A) less toxic metabolite
   B) metabolite more toxic to insects than mammals
   C) less polar metabolite
   D) metabolite more toxic to mammals than insects
   E) none of the above

5. Which of the following is most most likely to induce severe central nervous system depression and cardiac arrhythmia at toxic doses?
   A) arsenic
   B) aflatoxin
   C) naphthylamines
   D) carbon tetrachloride
   E) warfarin

6. The situation in which the combined effect of two chemicals is much greater than the sum of the effect of each agent when given alone is called
   A) potentiation
   B) addition
   C) synergism
   D) idiosyncrasy
   E) antagonism

7. Ethylene glycol
   A) is more toxic to man than to other animal species
   B) may produce calcium oxalate bladder stones
   C) may cause severe injury to renal tubules
   D) toxicity may be treated with ethanol
   E) all of the above

8. 2,3,7,8-Tetrachlorodibenzo-p-dioxin (TCDD) has a plasma half-life in humans of 5-10 years. From this observation, which of the following is correct?
   A) TCDD is so lipophilic that the fat tissue time constant is 5-10 years
   B) the half-life does not tell us anything directly about the fat tissue time constant
   C) TCDD is eliminated by metabolism which is slow
   D) the fat time constant is large and metabolism is slow
   E) TCDD is tightly bound to plasma protein

9. A body found in a factory is determined to have 20% carboxyhemoglobin. A compound that may be responsible for the fatality is
   A) diethyl ether
   B) liquid graphite
   C) propanol
   D) methylene chloride
   E) hydrogen cyanide

10. Half-lives of chemical substances are frequently reported in toxicokinetic studies. Which statement about the half-life is NOT correct?
   A) the half-life is the time required for the concentration of a substance to decline to half its initial value
   B) half-lives are frequently estimated from semilogarithmic plots of concentration versus time
   C) complete elimination of a substance from a biological system takes two half-lives
   D) half-lives may vary when elimination processes are concentration-dependent (e.g., saturable metabolism)
   E) half-lives for a specific chemical substance may be very different in different species of animals

11. The amount of alcohol in blood can be determined by taking an expired air sample. This is acceptable under the law because
   A) the blood/air solubility coefficient for ethanol is very low
   B) expired air is close to being in equilibrium with pulmonary capillary blood
   C) expired air represents the fraction of ethanol not metabolized
   D) it is easier to measure ethanol in air than in blood
   E) ethanol in blood varies much more than ethanol in expired air

12. A young man was admitted to an emergency room in a deep coma. His friends state that he took an "I.V. fix" of drug A six hours earlier. Therapeutic drug monitoring reveals a serum concentration of 0.5 mg/L. Given the toxicokinetic parameters provided below for this person, how much drug A did he inject six hours ago? (Vd = 100L, Cl = 51.2 L/hour, half-life = 3 hours).
   A) 25 mg
   B) 50 mg
   C) 100 mg
   D) 200 mg
   E) 400 mg

13. Which of the following laboratory animal species require dietary vitamin C?
   A) rat and mouse
   B) guinea pig and rabbit
   C) dog and monkey
   D) monkey and guinea pig
   E) rabbit and chicken

14. In the major air pollution catastrophe that occurred in Italy in 1976, the toxicant of greatest concern was
   A) hydrogen cyanide
   B) sulfur dioxide
   C) sulfur trioxide
   D) 1,2,7,8-tetrachlorodibenzo-p-dioxin
   E) 2,4,5-trichlorophenoxyacetic acid

15. Which of the following epidemiologic pairings is INCORRECT?
   A) incidence rate - number of healthy people in a population who develop a disease over a specified time
   B) prevalence rate - number of people in a population who have a particular disease at a given time
   C) observer variation - variability in finding due to variation among observers
   D) systematic error - fluctuations around a true value because of sampling error
   E) measurement error in independent variable - reduction of regression line slope

16. In general, the organ systems of newborn rodents are
   A) very similar to human newborns with respect to development
   B) more developed than human newborns
   C) much less sensitive to toxicants than human newborns
   D) much more sensitive to toxicants than human newborns
   E) less developed than human newborns

17. Arochlor 1254
   A) contains 54% by weight of biphenyl
   B) is formed from burning teflon
   C) has 54% of all possible sites occupied by chlorine
   D) causes chloracne
   E) none of the above

18. Approximately what concentration of carbon monoxide in the inspired air is required to cause a 50% carboxyhemoglobinemia?
   A) 50%
   B) 21%
   C) 7%
   D) 0.21%
   E) 0.07%

19. Exposure to which of the following may be due to incomplete combustion of carbonaceous fuels as well as to cigarette smoking, and is a cause of elevated carboxyhemoglobin levels in blood
   A) carbon dioxide
   B) carbon monoxide
   C) carbaryl
   D) carbon
   E) carbon tetrachloride

20. Which of the following statements about botulism is NOT true?
   A) poisoning occurs in ducks eating vegetation decaying in the anaerobic conditions of stagnant pools of water
   B) a postmortem diagnosis of botulism can be easily made since diagnostic lesions are present in liver and brain
   C) botulinum toxin prevents the release of acetylcholine at the neuromuscular junction
   D) the classic cause of death in botulism is the paralysis of the respiratory muscles
   E) immunization of guinea pigs with a specific antitoxin will protect the animals from experimental induction of botulism

21. When comparing the distribution of the population of sample means with the distribution of the population from which the samples were drawn, they will be most alike in which of the following parameters?
A) their size (n)
B) their means
C) their variance
D) their standard deviations
E) their correlation coefficients

22. Which statement best describes a "prospective cohort study" of the effects of a chemical exposure?
A) has the advantage over retrospective (historical) cohort study of being short-term with relatively few confounding factors
B) involves comparison of a diseased cohort with a sample of a non-diseased cohort
C) involves individuals, healthy at the beginning of the study, whose health and exposure are studied over time
D) uses predetermined observations in examining the health records of deceased individuals with known exposure histories
E) is the least desirable means of measuring changes in health parameters and relating them to exposure

23. Which of the following statements regarding uncertainty factors is false?
A) a 10-fold uncertainty factor is often used to extrapolate from animals to humans
B) a 10-fold uncertainty factor is often used when extrapolating results from a subchronic exposure to a chronic exposure
C) a 100-fold uncertainty factor is often used to account for variability in responses of individual humans
D) 10-fold uncertainty factor is often used to extrapolate from a LOAEL to a NOAEL
E) uncertainty factors are assumed to act independently

24. Which of the following items is not a part of the quantitative baseline risk assessment procedure for a Superfund site?
A) dose response assessment
B) benefit-cost assessment
C) qualitative or quantitative uncertainty analysis
D) risk characterization
E) exposure assessment

25. Steroid hormones act on the liver to alter the function of
A) endoplasmic reticulum
B) mitochondria
C) lysosomes
D) peroxisomes
E) Golgi appartus

Key for the following questions
Choose A) if 1,2 and 3 are correct
Choose B) if 1 and 3 are correct
Choose C) if 2 and 4 are correct
Choose D) if only 4 is correct
Choose E) if all four answers are correct

26. Which of the following statements is/are correct?
1) fluoroacetate inhibits mitochondrial ATP production
2) retinal enzymes cannot oxidize formaldehyde
3) fluoroacetate toxicity may be antagonized by ethanol
4) methanol toxicity may be antagonized by ethanol

27. Which of the following is correct concerning use of alternatives to animal tests for U.S. regulatory purposes ?
1) the alternative test needs only to be published in a scientific journal
2) the use of animals is eliminated
3) acceptance by the European Union will satisfy U.S. regulatory agencies
4) the validation and acceptance criteria of the Interagency Coordinating Committee on the Validation of Alternative Methods must be met

28. The reference dose or RfD is
1) essentially the same as the acceptible daily intake (ADI)
2) calculated from the LOAEL using uncertainty and modifying factors
3) calculated from the NOEL using uncertainty and modifying factors
4) the lowest dose causing mortality, reduced by safety factors

29. The clearance of a chemical is an important pharmacokinetic parameter. Which of the following is/are correct?
1) clearance is a process in which a chemical is removed permanently from the circulation
2) clearance represents the volume of blood (or plasma) that is cleared of a chemical per unit time
3) the plasma clearance represents the sum of all individual clearance processes such as metabolism, renal and biliary excretion
4) for a first-order process, an increase in plasma concentration will lead to an increase in clearance

30. Which of the following differences between adults and newborns modify the potential adverse effects of a chemical?
1) degree of plasma protein binding
2) xenobiotic metabolizing enzyme activity
3) rates of biliary excretion
4) food and water intake

31. Differences in which factor(s) affect the rate of metabolism of foreign compounds?
1) species
2) strain
3) gender
4) circadian rhythm

32. Which of the following cause(s) toxicity at least in part due to formation of toxic oxygen radicals through the process of "redox cycling"?
1) cyanide
2) 1,2,3,6,-tetrahydro-1-methyl-phenyl-4-pyridine (MPTP)
3) methylene blue
4) paraquat

33. Which of the following statements is/are true for paraquat?
1) stimulates oxygen consumption
2) stimulates the oxidation of NADPH
3) stimulates the generation of superoxide anion free radicals
4) undergoes oxidative dehalogenation in aerobic solutions

34. Which of the following agents alter(s) the binding affinity of hemoglobin for oxygen?
1) hydrogen cyanide
2) sodium nitrite
3) nitrogen dioxide
4) carbon monoxide

35. In the U.S. several federal agencies are responsible for regulation of toxic chemicals. Which of the following is/are correct?
1) the EPA regulates pesticides, air pollutants, water pollutants and pesticide residues in meat and poultry
2) the FDA regulates human drugs, animal drugs, cosmetics and food additives
3) the CPSC regulates hazardous consumer products, toxic household products and labeling of hazardous consumer products
4) the OSHA regulates toxic chemicals in the workplace and the transport of hazardous materials (HAZMAT)

36. The Food Quality Protection Act of 1996 has introduced some important changes in the regulation of pesticides in the United States. Which of the following regarding the Act is/are true?
1) establishes a statutory definition of "safety" as a reasonable certainty of no harm
2) requires an additional ten-fold margin of safety, under certain circumstances, to protect infants and children
3) requires EPA to consider aggregate and cumulative exposure to other related substances with common mechanisms when establishing tolerances for any one pesticide
4) amends both the Federal Insecticide, Fungicide and Rodenticide Act and the Federal Food, Drug and Cosmetic Act by deleting the Delaney clause

37. Good Laboratory Practice testing programs are regulated by which of the following organizations?
1) U.S. EPA
2) OECD
3) U.S. FDA
4) ISO

38. Which of the following chemicals at toxic doses cause(s) cyanosis?
1) cyanide
2) carbon monoxide
3) rotenone
4) pentachlorophenol
5) arsenic

39. Which of the following is/are true?
1) metallic mercury is released from the earth's crust as a vapor
2) metallic mercury can be oxidized to inorganic divalent mercury
3) inorganic divalent mercury can be alkylated to dimethylmercury
4) inorganic divalent mercury can be reduced to metallic mercury

40. Which of the following is/are true concerning the health effects of polychlorinated biphenyls (PCBs) on wildlife? PCB's
1) interfere with reproduction in birds
2) induce microsomal enzymes
3) tend to bioaccumulate because of their lipophilicity and chemical stability
4) possess porphyrogenic action, estrogenic activity and are immunosuppressive

41. Which of the following statements is/are NOT correct regarding the Proportional Mortality Ratio (PMR)?
1) the PMR permits estimation of the proportion of lives saved by eradication or reduction of a given cause of death
2) the PMR is the ratio of observed deaths in a study population relative to the number of expected deaths in the study population
3) in occupational studies, the PMR is the number of deaths from a specific cause over a specified interval relative to the total deaths in the same interval
4) the PMR and SMR can be used interchangeably

42. ANOVA as it applies to biostatistics
1) is a statistical test for the comparison of data that are independent and normally distributed
2) is a statistical test which compares non-continuous data that are normally distributed but are dependent
3) refers to "analysis of variance"
4) is a statistical tool which can be used for the calculation of an MTD

43. When conducting studies on a product intended to be marketed as a household cleaner containing an antimicrobial ingredient, study protocols must follow the criteria defined for
1) approval by the Food and Drug Administration
2) approval by the Consumer Product Safety Commission
3) labeling under the Federal Hazardous Substance Act
4) registration under the Federal Insecticide, Fungicide and Rodenticide Act

44. Which of the following is/are correct regarding occupational asthma?
1) industrial agents associated with asthma can be classified into large molecular weight allergens and small molecular weight chemical haptens
2) chemicals used industrially and associated with asthma include isocyanates, platinum salts and nickel salts
3) asthma is often underdiagnosed or misdiagnosed. Proper diagnosis can be confirmed by spirometry.
4) asthma will probably develop spontaneously in 10-20% of all workers during their working lives

45. Which of the following chemicals cause(s) hypoxemia in exercising animals?
1) cyanide
2) carbon monoxide
3) rotenone
4) pentachlorophenol

46. Active transport of toxicants through biological membranes has which of the following characteristics?
1) movement against electrochemical gradients and/or concentration gradients
2) saturable at high substrate concentrations
3) requires energy
4) competitive inhibition between compounds that are co-transported

47. Perchlorate exposure can inhibit iodine uptake by thyroid. This inhibition may result in which of the following?
1) increased circulating T_4 and T_3
2) decreased circulating T_4 and T_3
3) decreased TSH production by the pituitary
4) increased TSH production by the pituitary

48. In recent years, endocrine disrupting chemicals have received a great deal of interest in the scientific community as well as the lay media. Which of the following statements is/are correct?
1) studies of wildlife populations throughout the U.S. have found a causal relationship between exposure to endocrine disrupting chemicals and adverse developmental and reproductive effects
2) organochlorine pesticides and PCBs are examples of chemicals with estrogenic activity
3) the endocrine disrupting chemicals appear to produce their effects by binding with DNA
4) phytoestrogens and mycotoxins are examples of naturally occurring endocrine disrupting substances

49. Which of the following is/are correct?
1) inorganic arsenic is more toxic than organic arsenic
2) trivalent arsenic is more toxic than pentavalent and zero valent arsenic
3) in chronic arsenic poisoning, Mee's lines (transverse, white striae of the fingernails) may be evident
4) arsenic exposure can be ascertained by analysis of hair samples

50. One of the better known genetic metabolic polymorphisms is the ability to acetylate certain compounds. Persons who are slow acetylators may be particularly sensitive to which of the following agents?
1) benzidine
2) hydralazine
3) isoniazid
4) primaquine